



A mathematical model for the nutrient distribution of a spheroidal avascular cancer tumour within an inhomogeneous environment

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Abstract

When a cancerous cell colony grows within a healthy environment, the entire structure can be modelled as a continuous two-phase fluid with five bounded compartments, governed by the laws of mass conservation, Fick's diffusion law, and fluid mechanics principles. The interfaces of the five bounded compartments of the colony are defined by critical values of nutrient concentration. In studying the evolution of the exterior tumour boundary, nutrient concentration is the primary parameter. Although most existing research focuses on spherical tumours, significant implications for nutrient distribution emerge when spherical symmetry is abandoned, such as the occurrence of critical values at specific points rather than across the entire surface. In this work, we consider an oblate spheroidal tumour and investigate the effects of non-homogeneity in both nutrient supply and consumption rates. Our findings indicate that critical values are encountered within the interior of a thin layer, rather than at a single interface, although the interface is still included. We study the variation of nutrient concentration on the tumour's interfaces through plots, highlighting the critical locations. The prolate spheroidal case can be derived via a simple transformation, and comparisons with similar spherical models are also discussed.

Keywords Applied mathematics modelling · Avascular cancer tumour · Spheroidal geometry

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1 Introduction

The development of cancerous tumours within healthy tissue is an extremely complex phenomenon that has garnered significant multidisciplinary scientific interest over the past century. Deepening our understanding of this phenomenon could potentially lead to better control and ultimately the elimination of its life-threatening effects. A cancerous tumour is essentially a colony of cells that grows by invading healthy host tissue. These cells consume nutrients and proliferate repeatedly before they die either through programmed cell death (apoptosis) or due to a lack of nutrients (necrosis). As the tumour grows, numerous interrelated processes occur, including nutrient diffusion from the surrounding tissue into the tumour, cell proliferation when nutrients are sufficient, growth inhibition when nutrients are scarce or inhibitory substances are present, cell death or disintegration, elastic interactions between the tumour tissue and the healthy tissue, and internal pressure effects.

Deterministic mathematical modelling has provided significant insights into understanding these processes, based on the experimental data available [1–4]. Experimental evidence indicates that a newborn tumour experiences an initial phase of exponential growth, followed by a growth delay, and finally a phase of linear growth until it reaches a steady state, where cell birth is balanced by cell death. The stability of the final tumour configuration is of great interest, as the tumour's ability to transition from an avascular to a vascularized state, where it can become benign or malignant, depends on its capacity to stimulate the growth of blood vessels. In studying the evolution of the exterior tumour boundary in this avascular state, nutrient concentration is the primary parameter. Therefore, it is crucial to have an accurate mathematical model to determine the nutrient distribution within the tumour structure.

The majority of mathematical modelling of the avascular stage in tumour growth deals with spherically symmetric tumours grown *in vitro* [3–7]. This choice is well justified, as in the steady state, the avascular tumour is typically contained within a sphere with a diameter of approximately 2 mm. At such a small scale, deviations from spherical symmetry are not considered quantitatively significant. However, both analytical and experimental works [8, 9] have shown that non-symmetric tumours can occur in confined surroundings due to the pressure effects on tumour cell motility. In these cases, qualitative features emerge in the nutrient concentration distribution within the avascular tumour that are not present in spherically symmetric growth. Such features are reported in studies on ellipsoidal tumours [8, 10, 11] and in studies where non-homogeneous nutrient supply is imposed [12–14]. In all cases, the regions of the tumour occupied by cells at different vitality phases cannot be distinguished by interfaces of constant nutrient values, unlike in homogeneous spherically symmetric models, where cell vitality is directly connected to the nutrient levels of the surrounding environment.

In order to study the nutrient distribution in the interior of an anisotropic avascular tumour, we make certain basic assumptions, following well-established references [2–7], listed below. The tumour is considered as an incompressible fluid, which receives nutrients by diffusion from its surrounding. The dead cell debris, as well as the cell metabolic products, inhibits proliferation, when they exceed certain critical levels. A fully developed tumour consists of four regions, formed with respect to the

nutrient and the internally produced inhibitor concentration levels. The dead cell debris form a necrotic core covered by a layer occupied by quiescent cells that do not have enough nutrient to proliferate. In the next layer, the cells are also quiescent, but this is due to high inhibitory levels, even though the nutrient present is adequate to support proliferation. In the exterior tumour layer the nutrient concentration is high enough and the inhibitor concentration is low enough so that the cells there proliferate. All parts of the fully developed tumour are characterized by the same diffusion constant. The tumour and its surroundings are in diffusive equilibrium at all times, as the diffusion time scale is significantly shorter than the growth time scale. The vitality state of the cells is controlled by the amount of nutrient which is available. The following critical nutrient values are considered. If $\sigma(\mathbf{r})$ denote the nutrient concentration at every point $\mathbf{r}$, then the minimum value of σ that can support proliferation is denoted by σ_p^* . The minimum value of σ that can support life, even in a quiescent state, is denoted by σ_n^* and in regions with nutrient concentration below that value only necrotic cells can be met. The nutrient is provided by the tumour's surrounding at concentration $\sigma_\infty(\mathbf{r}_e)$ on the exterior tumour's surface for $\mathbf{r} = \mathbf{r}_e$. Finally, the geometrical anisotropy is considered to be imposed on the tumour by the elastic properties of the surrounding.

In all studies where spherical geometry is used to model the tumour, the interfaces between tumour regions are characterized by constant nutrient concentration values [3–7], as the nutrient is proven to depend only on the radial position. In [8, 10, 11], where ellipsoidal modelling of the tumour is reported, the interior nutrient distribution is point dependent, and the tumour interfaces are no longer defined as the locus of cells with constant nutrient concentration values. In the present work, we closely examine the nutrient distribution within the asym-metrical tumour, assuming a more detailed model in the oblate spheroidal geometry [15], obtaining the field in terms of the relative eigenfunctions [16] via mathematical techniques, which are used in several physical sciences such as fluid dynamics [17, 18], magnetohydrodynamics [19], electromagnetism [20, 21] etc. We identify locations, where critical values are met and we propose an alternative definition of these critical values with respect to such geometry. The analytical results, numerically implemented, exhibit the expected behaviour for nutrient concentration. The corresponding model for a prolate spheroidal tumour can be addressed similarly, though it is not presented here. The related results are obtained by transforming from the oblate to the prolate spheroidal case, while our findings straightforwardly reduce to the well-known spherical tumour model.

The remainder of the paper is organized as follows. In Sect. 2, we formulate the problem in oblate spheroidal terms, and we give the solution of the nutrient concentration with the corresponding two-dimensional and three-dimensional plots, illustrating the distribution in the interior of the tumour structure. In Sect. 3, a discussion on the critical values is included, and in Sect. 4, the analytical transformation formulae to the prolate spheroidal case is provided and the reduction to the simpler spherical model is carried out. Finally, Sect. 5 is devoted to a discussion that summarizes the outcomes of the current research, giving slight hints to further analysis and development of the model.

2 The nutrient concentration problem

We consider a fully developed avascular tumour that grows maintaining all its boundary surfaces as confocal oblate spheroids. Therefore, we consider an oblate spheroidal system [15], defined by the semifocal distance $c > 0$, and we use the transformed oblate spheroidal coordinates (λ, ζ, ϕ) , which define every position $\mathbf{r}$ inside the tumour and are connected to the Cartesian coordinates (x_1, x_2, x_3) via the equations:

$$\begin{aligned}x_1 &= c \sqrt{1 + \lambda^2} \sqrt{1 - \zeta^2} \cos \phi, \\x_2 &= c \sqrt{1 + \lambda^2} \sqrt{1 - \zeta^2} \sin \phi, \\x_3 &= c \lambda \zeta,\end{aligned}\quad (1)$$

where the corresponding variables are $\lambda \geq 0$, $\zeta \in [-1, 1]$ and $\phi \in [0, 2\pi)$. In such terms, the tumour consists of a necrotic core, $\Omega_n = \{(\lambda, \zeta, \phi) \in \mathbb{R}^3 : 0 \leq \lambda < \lambda_n, -1 \leq \zeta \leq 1, 0 \leq \phi < 2\pi\}$, where $0 < \sigma < \sigma_n^*$, surrounded by a quiescent layer $\Omega_q = \{(\lambda, \zeta, \phi) \in \mathbb{R}^3 : \lambda_n < \lambda < \lambda_q, -1 \leq \zeta \leq 1, 0 \leq \phi < 2\pi\}$, occupied by hypoxic cells, which are alive, but are not able to proliferate, since it holds that $0 < \sigma_n^* < \sigma < \sigma_p^*$. In the subsequent layer, namely $\Omega_{p-} = \{(\lambda, \zeta, \phi) \in \mathbb{R}^3 : \lambda_q < \lambda < \lambda_{p-}, -1 \leq \zeta \leq 1, 0 \leq \phi < 2\pi\}$, even though we have $\sigma > \sigma_p^*$, which means that there is enough nutrient to support proliferation, at the same time, there is too much inhibitor concentration present to allow division. Nevertheless, in the next layer $\Omega_{p+} = \{(\lambda, \zeta, \phi) \in \mathbb{R}^3 : \lambda_{p-} < \lambda < \lambda_{p+}, -1 \leq \zeta \leq 1, 0 \leq \phi < 2\pi\}$, cell proliferation takes place, since $\sigma > \sigma_p^*$, while the inhibitor concentration is reduced significantly. Finally, the compartment of the normal host tissue $\Omega_e = \{(\lambda, \zeta, \phi) \in \mathbb{R}^3 : \lambda_{p+} < \lambda < \lambda_e, -1 \leq \zeta \leq 1, 0 \leq \phi < 2\pi\}$ is affected by the cancerous mass growth. Let S_j , $j = n, q, p-, p+, e$, denote the outer surface of each oblate spheroidal region. On the exterior surface S_e , the surrounding provides nutrient in the form $\sigma_\infty(\mathbf{r}_e)$. The oblate spheroidal shells, described earlier, are shown in Fig. 1.

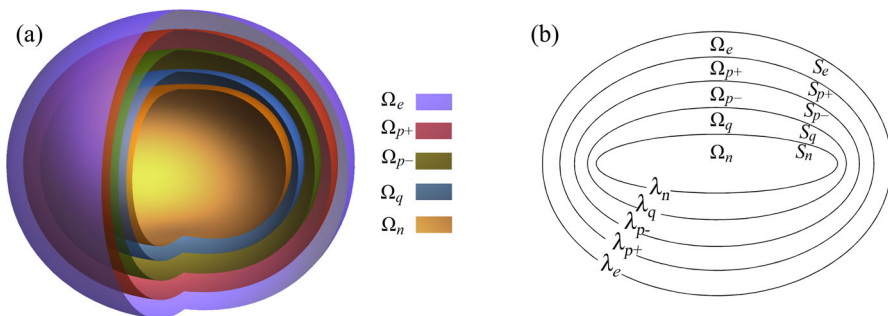


Fig. 1 **a** The oblate spheroidal compartments Ω_j for $j = n, q, p-, p+, e$ and **b** the sequence of elliptical cross sections as plots of lines of constant $\lambda = \{\lambda_n, \lambda_q, \lambda_{p-}, \lambda_{p+}, \lambda_e\}$ in the x_1x_3 -plane

The nutrient diffuses inward to the tumour and it is consumed at a consumption rate that depends on the vital state of the cells and reflects the corresponding phase of the cell cycle. In particular, we denote by γ_j the consumption rate normalized by the diffusion constant k_σ at the corresponding region Ω_j , $j = n, q, p-, p+, e$ and demand that $\gamma_n = 0$ as Ω_n is occupied by necrotic debris and $\gamma_q = \gamma_{p-}$ (see [4] for instance), since the cells in both Ω_q and Ω_{p-} are quiescent, but they are also capable of proliferation, which evidently is not achieved due to the presence of an inhibitor. On the other hand, $\gamma_{p+} := \gamma_p \geq \gamma_q$ as the cells in Ω_{p+} are in a more active proliferating state and, finally, $\gamma_e \geq \gamma_q$ but γ_e can be either greater or lower than γ_p depending on the kind of the healthy tissue in which the tumour grows. Probably, the suggestion of $\gamma_e \leq \gamma_p$ has a physical reasoning due to the greater demands of the much longer proliferation phase of the cancerous cell cycle compared to the normal cell cycle [4], which means that a shorter cell cycle for tumour cells can also be interpreted as greater nutrient consumption ($\gamma_e \leq \gamma_p$). In addition, we assume that as long as the cell remains in a certain phase of its cycle, its needs in nutrients are unaltered, no matter the availability on nutrients. Therefore, we suggest that all cells occupying the same tumour region have the same constant consumption rate, while the transmission from one region to another results to a steep and continuous change [22] in the consumption rate. These considerations lead to modelling the nutrient consumption rate, where the desired behaviour is approximated by the functional

$$\gamma(\lambda; h) = \begin{cases} \gamma_e, & \lambda_{p+} \leq \lambda < \lambda_e \\ \gamma_p, & \lambda_{p-,h} \leq \lambda < \lambda_{p+} \\ \gamma_q + (\gamma_p - \gamma_q)f(\lambda - \lambda_{p-}; h), & \lambda_{n,h} \leq \lambda < \lambda_{p-,h} \\ \gamma_q f(\lambda - \lambda_n; h), & 0 \leq \lambda < \lambda_{n,h} \end{cases}$$

with $\lambda_{n,h} := \lambda_n + \frac{\lambda_q - \lambda_n}{2}$ and $\lambda_{p-,h} := \lambda_{p-} + \frac{\lambda_{p+} - \lambda_{p-}}{2}$, (2)

where $f(x; h) = \frac{1}{2} + \frac{1}{2} \tan h\left(\frac{x}{h}\right)$ and h is a very small positive parameter $0 < h \ll 1$ that controls the steepness of the drop. It is readily seen that for $\lambda = \lambda_n$, $\gamma = \gamma_{q,h} := \frac{\gamma_q}{2}$, while for $\lambda = \lambda_{p-}$, $\gamma = \gamma_{p,h} := \gamma_q + \frac{\gamma_p - \gamma_q}{2}$. Functional (2) is sketched in Fig. 2 for specific values of λ_j , $j = n, q, p-, p+, e$ as well γ_q , γ_p and γ_e , whereas we consider the question on the significance of the jump discontinuities of the nutrient flow across S_n and S_{p-} that are implied, by modelling the steep drop between the consumption rates in the different regions with function (2). The main role of functional (2) is to motivate the jump conditions in the sequel, wherein an appropriate model yields the limit as h tends to zero. When $h \rightarrow 0$, then $\lambda_{n,h} \rightarrow \lambda_n$, $\lambda_{p-,h} \rightarrow \lambda_{p-}$ and the modified function f becomes the known Heaviside function H [22], i.e. $\lim_{h \rightarrow 0} f(x; h) = H(x) =$

$$\begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}. \text{ Therein, function } \gamma \text{ takes the sectional continuous form:}$$

$$\gamma(\lambda) = [H(\lambda - \lambda_n) - H(\lambda - \lambda_{p-})]\gamma_q + [H(\lambda - \lambda_{p-}) - H(\lambda - \lambda_{p+})]\gamma_p + [H(\lambda - \lambda_{p+}) - H(\lambda - \lambda_e)]\gamma_e, \quad (3)$$

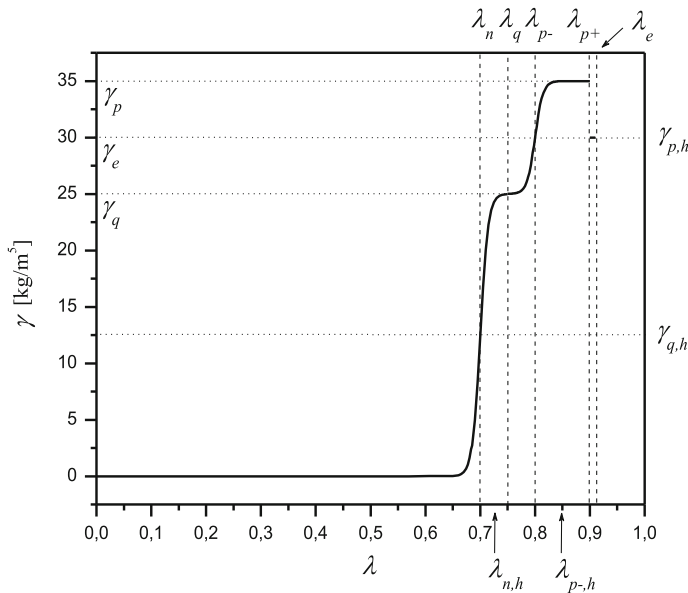


Fig. 2 Variation of the functional $\gamma(\lambda; h)$, $h = 0.003$ for oblate parameters $\lambda_n = 0.7$, $\lambda_q = 0.75$, $\lambda_{p-} = 0.8$, $\lambda_{p+} = 0.9$, $\lambda_e = 0.91$ and for normalized consumption rates $\gamma_q = 25 \text{ kg/m}^5$, $\gamma_p = 35 \text{ kg/m}^5$, $\gamma_e = 30 \text{ kg/m}^5$

which was the expression adopted in [12] for the particular case $\gamma_e = 0$.

As shown in detail in [12], the nutrient flow is only diffusive with constant k_σ , hence, it follows the Fick law and it is directed to regions of lower nutrient concentration with respect to the diffusive equation $\partial\sigma/\partial t - k_\sigma\Delta\sigma + \Gamma_j = 0$ with $\sigma = \sigma(\mathbf{r}, t)$, where $\Gamma_j = k_\sigma\gamma_j$ for $j = n, q, p-, p+, e$ are the corresponding regular and without normalization consumption rates. Since the nutrient is in a diffusive equilibrium state, there is no time dependence and it is described with elliptic instead of parabolic partial differential equations. Therefore, the boundary value problems for very small values for the parameter h are proved to become

$$\Delta\sigma(\mathbf{r}) = \gamma_j \text{ for } j = n, q, p-, p+, e \quad (4)$$

with continuity conditions for σ on every inner boundary S_n, S_q, S_{p-} and S_{p+} . On S_q and S_{p+} the normal nutrient flow is continuous, which reads as $\hat{\lambda} \cdot \nabla\sigma_{p-}(\mathbf{r}_q) - \hat{\lambda} \cdot \nabla\sigma_q(\mathbf{r}_q) = 0$ and $\hat{\lambda} \cdot \nabla\sigma_e(\mathbf{r}_{p+}) - \hat{\lambda} \cdot \nabla\sigma_{p+}(\mathbf{r}_{p+}) = 0$, respectively, where $\hat{\lambda}$ is the outward unit normal vector on the corresponding surface. By applying the oblate spheroidal gradient operator [15] to obtain

$$\hat{\lambda} \cdot \nabla = \frac{\sqrt{1+\lambda^2}}{c\sqrt{\lambda^2+\zeta^2}} \frac{\partial}{\partial\lambda} \text{ for every } \lambda \in [0, \lambda_e] \text{ and } \zeta \in [-1, 1], \quad (5)$$

the boundary condition on S_q assumes the form:

$$\frac{\partial \sigma_{p-}(\lambda_q, \zeta, \phi)}{\partial \lambda} - \frac{\partial \sigma_q(\lambda_q, \zeta, \phi)}{\partial \lambda} = 0 \text{ for every } \zeta \in [-1, 1] \text{ and } \phi \in [0, 2\pi) \quad (6)$$

and, similarly, on S_{p+} it holds

$$\frac{\partial \sigma_e(\lambda_{p+}, \zeta, \phi)}{\partial \lambda} - \frac{\partial \sigma_{p+}(\lambda_{p+}, \zeta, \phi)}{\partial \lambda} = 0 \text{ for every } \zeta \in [-1, 1] \text{ and } \phi \in [0, 2\pi). \quad (7)$$

On the other hand, the function $f(x; h)$ results to a jump discontinuity of the normal derivative of σ across the interfaces S_n and S_{p-} as for infinitesimally small values of the parameter h , the function $f(x; h)$ becomes the step function $H(x)$, whose derivative with respect to variable x provides with a delta sequence [23], that is $H'(x) = \delta(x)$ and acts as a generalized function. Under some extensive but careful analysis, which is based on the integration via a limiting procedure (see [12] for technical details), the corresponding boundary conditions become $\hat{\lambda} \cdot \nabla \sigma_q(\lambda_n, \zeta, \phi) - \hat{\lambda} \cdot \nabla \sigma_n(\lambda_n, \zeta, \phi) = c\gamma_q \sqrt{\lambda_n^2 + \zeta^2} / 2\sqrt{1 + \lambda_n^2}$ on S_n or, by virtue of (5), it is

$$\frac{\partial \sigma_q(\lambda_n, \zeta, \phi)}{\partial \lambda} - \frac{\partial \sigma_n(\lambda_n, \zeta, \phi)}{\partial \lambda} = \frac{c^2 \gamma_q \lambda_n^2 + \zeta^2}{2(1 + \lambda_n^2)} \text{ for every } \zeta \in [-1, 1] \text{ and } \phi \in [0, 2\pi) \quad (8)$$

and, respectively, $\hat{\lambda} \cdot \nabla \sigma_{p+}(\lambda_{p-}, \zeta, \phi) - \hat{\lambda} \cdot \nabla \sigma_{p-}(\lambda_{p-}, \zeta, \phi) = c(\gamma_p - \gamma_q) \sqrt{\lambda_{p-}^2 + \zeta^2} / 2\sqrt{1 + \lambda_{p-}^2}$ on S_{p-} or, by means of (5), it holds

$$\frac{\partial \sigma_{p+}(\lambda_{p-}, \zeta, \phi)}{\partial \lambda} - \frac{\partial \sigma_{p-}(\lambda_{p-}, \zeta, \phi)}{\partial \lambda} = \frac{c^2 (\gamma_p - \gamma_q) \lambda_{p-}^2 + \zeta^2}{2(1 + \lambda_{p-}^2)} \text{ for every } \zeta \in [-1, 1] \text{ and } \phi \in [0, 2\pi). \quad (9)$$

Finally, on the exterior surface S_e , the nutrient field $\sigma_\infty(\mathbf{r}_e)$ is assumed to be supplied in the form of infinite series in terms of the Legendre functions of the first kind P_ℓ for $\ell \geq 0$ [16], i.e.

$$\sigma_\infty(\mathbf{r}_e) = \sum_{\ell=0}^{\infty} \sigma_{\infty, \ell}(\lambda_e) P_\ell(\zeta) \text{ for every } \zeta \in [-1, 1]$$

$$\text{with } \sigma_{\infty, \ell}(\lambda_e) = \frac{2\ell + 1}{2} \int_{-1}^{+1} \sigma_\infty(\mathbf{r}_e) P_\ell(\zeta) d\zeta \text{ for } \ell \geq 0, \quad (10)$$

wherein any other smooth function for $\sigma_\infty(\mathbf{r}_e)$ accepts an expansion of the form (10), since P_ℓ for $\ell \geq 0$ are orthogonal with respect to the classical inner product of the

Legendre functions of the first kind (e.g. see [16] for more details). Here, it is important to mention that the nutrient concentration field has to be regular at the origin, while on the boundaries S_n and S_q , the critical values are met.

Applying a standard method of separation of variables [15, 16] in every compartment Ω_j for $j = n, q, p-, p+, e$ and performing tedious but straightforward calculations based on the proper application of the aforementioned boundary conditions, the nutrient field is obtained in a compact fashion as follows:

$$\begin{aligned} \sigma(\lambda, \zeta, \phi) = & \frac{c^2}{9} \left[H(\lambda - \lambda_n) \gamma_q + H(\lambda - \lambda_{p-}) (\gamma_p - \gamma_q) + H(\lambda - \lambda_{p+}) (\gamma_e - \gamma_p) \right] \\ & \left[\frac{1}{2} - (P_2(i\lambda) + P_2(\zeta)) \right] \\ & + \frac{c^2}{9} \sum_{m=0,2} \left\{ \left[\delta_{0m} \left(P_2(i\lambda_e) - \frac{1}{2} \right) + \delta_{2m} \right] \gamma_e \frac{P_m(i\lambda)}{P_m(i\lambda_e)} \right. \\ & + \frac{3}{2} \sum_{j=n, p-, p+} \frac{\delta_{jn} \gamma_q + \delta_{jp-} (\gamma_p - \gamma_q) + \delta_{jp+} (\gamma_e - \gamma_p)}{P_m(i\lambda_j) Q'_m(i\lambda_j) - P'_m(i\lambda_j) Q_m(i\lambda_j)} \\ & \left[g_{m,j}^{(i)} (1 - H(\lambda - \lambda_j)) P_m(i\lambda) - g_{m,j}^{(e)} \right. \\ & \left. \left(H(\lambda - \lambda_j) (i Q_m(i\lambda)) - (i Q_m(i\lambda_e)) \frac{P_m(i\lambda)}{P_m(i\lambda_e)} \right) \right] \Big\} P_m(\zeta) \\ & + \sum_{\ell=0}^{\infty} \sigma_{\infty, \ell}(\lambda_e) \frac{P_{\ell}(i\lambda)}{P_{\ell}(i\lambda_e)} P_{\ell}(\zeta) \\ & \text{for every } \lambda \in [0, \lambda_e], \zeta \in [-1, 1] \text{ and } \phi \in [0, 2\pi), \end{aligned} \quad (11)$$

wherein we have used the limiting version of the consumption function (3). Within expression (11), the primed notation represents differentiation with respect to the appropriate argument, followed by substitution. P_{ℓ} and Q_{ℓ} for $\ell \geq 0$ are the Legendre functions of first and of the second kind, respectively [16] and H is the Heaviside function, defined earlier, while due to the rotational symmetry of the introduced boundary value problem (4)–(10), the nutrient concentration field is independent of the azimuthal angle $\phi \in [0, 2\pi)$. Moreover, $i = \sqrt{-1}$ is the imaginary unit inserted properly so as σ be real, δ_{ij} is the delta of Kronecker ($\delta_{ii} = 1$ and zero otherwise) and the following geometrical parameters appear in (11), those are

$$g_{0,j}^{(e)} = -g_{2,j}^{(e)} = -2\lambda_j + \frac{(3\lambda_j^2 + 1)(1 - \delta_{jp+})}{1 + \lambda_j^2} \text{ for } j = n, p-, p+, \quad (12)$$

$$g_{0,j}^{(i)} = \frac{3\lambda_j^2 + 2}{3(1 + \lambda_j^2)} + g_{0,j}^{(e)} \tan^{-1} \lambda_j \text{ for } j = n, p-, p+ \quad (13)$$

and

$$g_{2,j}^{(i)} = \frac{3\lambda_j^2 + 2}{3(1 + \lambda_j^2)} - \frac{3\lambda_j}{1 + \lambda_j^2}(1 - \delta_{jp+}) + \frac{1}{2}\left(1 + g_{2,j}^{(e)} \tan^{-1} \lambda_j\right) \text{ for } j = n, p-, p+. \quad (14)$$

In order to illustrate the analytical result (11), accompanied by (12)–(14), we perform the above analysis for the special case where the nutrient concentration is supplied from the host surrounding in a form that enhances the oblate spheroidal structure, namely, when $\sigma_{\infty, \ell}(\lambda_e) = 0$ for $\ell \in \mathbb{N} - \{0, 2\}$. In addition, we retain the very same considerations for λ_j , $j = n, q, p-, p+, e$ and $\gamma_q, \gamma_p, \gamma_e$ to plot $\sigma(\lambda, \zeta, \phi)$ for every $\lambda \in [0, \lambda_e]$, $\zeta \in [-1, 1]$ and $\phi \in [0, 2\pi)$ via (11). These geometrical and physical characteristic values of the tumour are estimated on the basis of [2–6, 9] and mostly [14], wherein the approximate size of the tumour and the corresponding physical parameters are approximated sufficiently and conform with physical reality.

To that purpose, Fig. 3 depicts the variation of σ with respect to λ for several values of the angular variable ζ . We observe that σ is a continuous function, where the effect of (8) and (9) at $\lambda = \lambda_n$ and $\lambda = \lambda_{p-}$, respectively, is shown, resulting to different inclinations, while for $\lambda = \lambda_n$, one obtains σ_n^* , while on $\lambda = \lambda_q$, the critical value σ_p^* is met. Also, as ζ increases from 0 to 1, we obtain the maximum value at $\zeta = 1$ and the minimum value is taken for $\zeta = 0.5$.

Figure 4 represents a contour plot of the nutrient concentration, denoted by σ

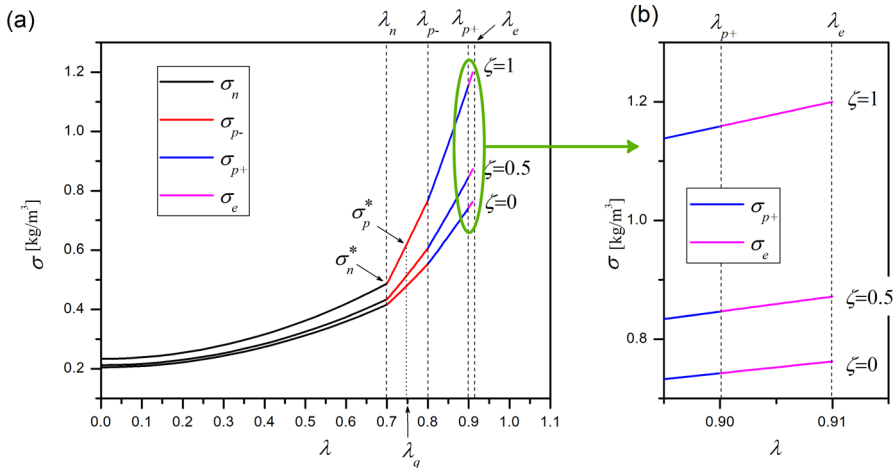
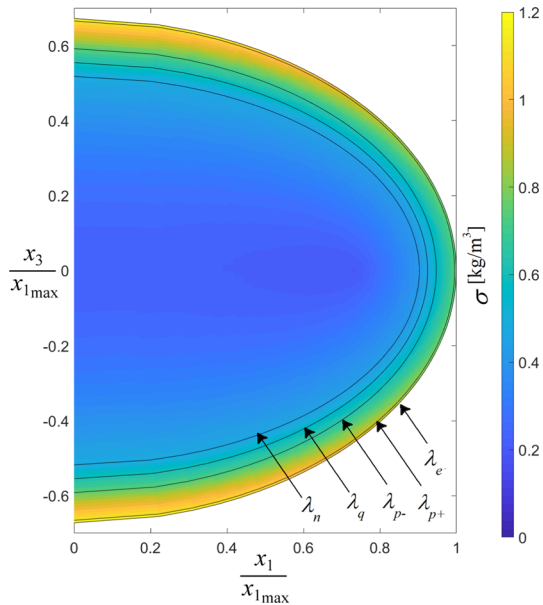


Fig. 3 **a** Variation of the nutrient concentration field $\sigma(\lambda, \zeta, \phi)$ at $\zeta = 0, 0.5, 1$ and for every $\lambda \in [0, \lambda_e]$, $\phi \in [0, 2\pi)$ for oblate parameters $\lambda_n = 0.7$, $\lambda_q = 0.75$, $\lambda_{p-} = 0.8$, $\lambda_{p+} = 0.9$, $\lambda_e = 0.91$, for normalized consumption rates $\gamma_q = 25 \text{ kg/m}^5$, $\gamma_p = 35 \text{ kg/m}^5$, $\gamma_e = 30 \text{ kg/m}^5$ and for nutrient supply $\sigma_{\infty, 0}(\lambda_e) = 0.375 \text{ kg/m}^3$, $\sigma_{\infty, 2}(\lambda_e) = 0.825 \text{ kg/m}^3$. **b** Magnification of the non-visible plotted area for the solution for $\lambda_{p+} < \lambda < \lambda_e$

Fig. 4 Variation of the nutrient concentration $\sigma(\lambda, \zeta, \phi)$ for every $\lambda \in [0, \lambda_e]$, $\zeta \in [-1, 1]$ and $\phi \in [0, 2\pi)$ for oblate parameters $\lambda_n = 0.7$, $\lambda_q = 0.75$, $\lambda_{p-} = 0.8$, $\lambda_{p+} = 0.9$, $\lambda_e = 0.91$, for normalized consumption rates $\gamma_q = 25 \text{ kg/m}^5$, $\gamma_p = 35 \text{ kg/m}^5$, $\gamma_e = 30 \text{ kg/m}^5$ and for $\sigma_{\infty,0}$ (λ_e) = 0.375 kg/m^3 , $\sigma_{\infty,2}$ (λ_e) = 0.825 kg/m^3



(kg/m^3), within a cross section of a fixed oblate spheroidal tumour geometry. The coordinates $x_1/x_{1\max}$ and $x_3/x_{1\max}$ are normalized spatial variables indicating the relative position within the tumour structure. The plot effectively demonstrates the distribution of nutrients within the tumour's oblate spheroidal geometry. The highest nutrient concentrations are found near the tumour's surface, decreasing as one moves inward towards the necrotic core. This distribution is crucial for understanding how different regions within the tumour are affected by nutrient availability, influencing cell phenotypes and their potential for proliferation or quiescence.

The contour plots of Fig. 5 show the variation of the nutrient concentration σ (kg/m^3) for different oblate parameters λ_n , λ_q , λ_{p-} , while all other values remain unaltered. The plots effectively demonstrate the distribution of nutrients within the tumour's oblate spheroidal geometry under two different scenarios. In Fig. 5a, the nutrient values encountered in the necrotic core are decreased with respect to the previous case (Fig. 4), while in Fig. 5b, the nutrient values are increased with respect to both previous cases. This observation highlights the relationship between the nutrient distribution and the size of the compartments.

3 The critical nutrient values

As mentioned in the introduction, there are two nutrient values critical for the growth of the tumour colony. The value σ_n^* determines whether a cell will starve out to death or it will survive and continue with its cell cycle and consequently it is expected to be met upon the interface S_n . Similarly the critical value σ_p^* is the minimum nutrient concentration demanded for the cell to be able to proliferate, so it is expected to be

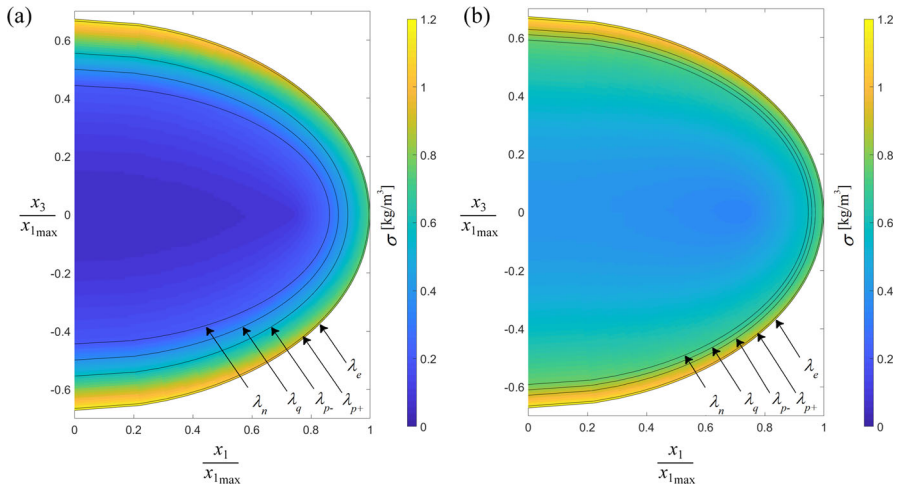


Fig. 5 Variation of the nutrient concentration $\sigma(\lambda, \zeta, \phi)$ for every $\lambda \in [0, \lambda_e]$, $\zeta \in [-1, 1]$ and $\phi \in [0, 2\pi]$ for oblate parameters **a** $\lambda_n = 0.6$, $\lambda_q = 0.675$, $\lambda_{p-} = 0.75$, $\lambda_{p+} = 0.9$, $\lambda_e = 0.91$ and **b** $\lambda_n = 0.8$, $\lambda_q = 0.825$, $\lambda_{p-} = 0.85$, $\lambda_{p+} = 0.9$, $\lambda_e = 0.91$, for normalized consumption rates $\gamma_q = 25 \text{ kg/m}^5$, $\gamma_p = 35 \text{ kg/m}^5$, $\gamma_e = 30 \text{ kg/m}^5$ and for $\sigma_{\infty,0}(\lambda_e) = 0.375 \text{ kg/m}^3$, $\sigma_{\infty,2}(\lambda_e) = 0.825 \text{ kg/m}^3$

met upon S_q . Actually, this is the case in a radially symmetric tumour colony, and therefore, in the literature, only this case is reported. However, when the symmetry is abandoned by assuming a non-symmetrical tumour, the model cannot fully provide this coincidence of the geometrical and physical boundary between the necrotic, quiescent and proliferating region, as shown also in [8, 12]. From (11) it is revealed that even for a given value of the radial variable λ , the nutrient concentration is not constant, but varies with the angular variable.

Since σ is a continuous function defined in the interior of the Ω_j compact set for any $j = n, q, p-, p+, e$, it takes a maximum and a minimum value on the boundary $\partial\Omega_j$. Therefore, let

$$\sigma_n^* = \min_{\substack{\zeta \in [-1, 1] \\ \phi \in [0, 2\pi]}} \sigma(\lambda_n, \zeta, \phi) \text{ for every } \zeta \in [-1, 1] \text{ and } \phi \in [0, 2\pi]. \quad (15)$$

On the other hand, due to the fact that the nutrient concentration σ is a continuous and increasing function of λ , there is a value $\lambda_n^* \in (0, \lambda_n)$ for which $\sigma(\lambda_n^*, \zeta, \phi) = \sigma_n^*$ for $\zeta \in [-1, 1]$ and $\phi \in [0, 2\pi]$, while there exists $\lambda \in [\lambda_n^*, \lambda_n]$, such that

$$\sigma_n^* \leq \sigma(\lambda, \zeta, \phi) \leq \max_{\substack{\zeta \in [-1, 1] \\ \phi \in [0, 2\pi]}} \sigma(\lambda_n, \zeta, \phi). \quad (16)$$

Therefore, the region, which is occupied by cells that are dead out of starvation is the interior of the oblate spheroid $\lambda \in (0, \lambda_n^*)$, while in the spheroidal shell $\lambda \in [\lambda_n^*, \lambda_n]$ nutrient can be found in concentration that may support life, even though the shell

might still be occupied by apoptotic cell debris. Following similar argumentation, the critical value σ_p^* can be defined as follows:

$$\sigma_p^* = \min_{\substack{\zeta \in [-1, 1] \\ \phi \in [0, 2\pi)}} \sigma(\lambda_q, \zeta, \phi) \text{ for every } \zeta \in [-1, 1] \text{ and } \phi \in [0, 2\pi). \quad (17)$$

This definition results to the existence of a radial value $\lambda_q^* \in (\lambda_n, \lambda_q)$, such that $\sigma(\lambda_q^*, \zeta, \phi) = \sigma_p^*$ for $\zeta \in [-1, 1]$ and $\phi \in [0, 2\pi)$, while there are points of (λ, ζ, ϕ) in the oblate spheroidal shell $\lambda \in [\lambda_q^*, \lambda_q]$, where

$$\sigma_p^* \leq \sigma(\lambda, \zeta, \phi) \leq \max_{\substack{\zeta \in [-1, 1] \\ \phi \in [0, 2\pi)}} \sigma(\lambda_q, \zeta, \phi). \quad (18)$$

In other words, in the latter region, nutrients are met also in concentrations that may support proliferation and the exclusively quiescent region is the spheroidal shell $\lambda \in [\lambda_n, \lambda_q^*]$. All the above considerations are eliminated in spherical tumour models that concern homogeneous nutrient supply, as the angular dependence no longer exists and the physical and geometrical inner boundaries coincide.

Obviously, there are other ways to define the critical values. Another choice, addressed in previous works (e.g. see [8, 12], refers to them as the mean value of σ on the corresponding surface, namely

$$\sigma_n^* = \frac{1}{|S_n|} \int_{S_n} \sigma(\lambda_n, \zeta, \phi) \, ds \text{ and } \sigma_p^* = \frac{1}{|S_q|} \int_{S_q} \sigma(\lambda_q, \zeta, \phi) \, ds, \quad (19)$$

where $|S_n|$ and $|S_q|$ are the areas of the surfaces S_n and S_q , respectively. Although this choice also leads to the existence of a spheroidal shell in the interior of Ω_n and of Ω_q , where the vitality state of the cells occupying them is not directly related to the nutrient concentration. Therefore, such alternative definition does not provide new perspective on the issue under consideration.

4 The prolate spheroidal and the spherical models

Herein, we wish to demonstrate the effectiveness of the aforementioned avascular tumour spheroidal model in an analytical manner by initially providing the necessary conditions, according to which the oblate spheroidal model is converted to the prolate spheroidal model. This is accomplished via the transformation [15]

$$\lambda \rightarrow -i\tau \text{ and } c \rightarrow i\bar{c}, \quad (20)$$

where $\tau \in [1, +\infty)$ and $\bar{c} > 0$ are the new characteristic variables of the prolate spheroidal geometry, rendering the corresponding results in a simple and elegant way, since there do not appear any indeterminacies.

In the sequel, using the standard reduction relations [15], the oblate spheroidal geometry degenerates to the spherical one (r, ζ, ϕ) , $r \in [0, +\infty)$ and ζ, ϕ as usual, in the limit as $c \rightarrow 0$. At that limit, we have $\lim_{c \rightarrow 0} (c\lambda) = r$, resulting to

$$\lim_{c \rightarrow 0} [(ic)^\ell P_\ell(i\lambda)] = \frac{(2\ell)!}{2^\ell (\ell!)^2} r^\ell \text{ for every } r \in [0, +\infty) \quad (21)$$

and

$$\begin{aligned} & \lim_{c \rightarrow 0} [(ic)^{-(\ell+1)} Q_\ell(i\lambda)] \\ &= \left\{ \frac{1}{2^\ell} \sum_{k=0}^{[\ell/2]} \frac{(-1)^k (2\ell - 2k)!}{k! (\ell - k)! (\ell - 2k)! (2\ell - 2k + 1)!} \right\} \frac{1}{r^{\ell+1}} \text{ for every } r \in [0, +\infty), \end{aligned} \quad (22)$$

wherein $[\ell/2]$ is the integer part of $\ell/2$ for $\ell \in \mathbb{N}$. Therefore, a cumbersome mathematical treatment upon the nutrient result (11) leads to the corresponding spherical expression, given by

$$\begin{aligned} \lim_{c \rightarrow 0} \sigma(\lambda, \zeta, \phi) := \sigma(r, \zeta, \phi) &= \frac{1}{6} \left\{ [H(\lambda - \lambda_n) \gamma_q + H(\lambda - \lambda_{p-}) (\gamma_p - \gamma_q) \right. \\ &\quad \left. + H(\lambda - \lambda_{p+}) (\gamma_e - \gamma_p)] r^2 + \gamma_e r_e^2 \right\} \\ &+ \frac{1}{6} \sum_{j=n, p-p+} [\delta_{jn} \gamma_q + \delta_{jp-} (\gamma_p - \gamma_q) + \delta_{jp+} (\gamma_e - \gamma_p)] r_j^2 \\ &\left[2H(\lambda - \lambda_j) - 2r_j \left(H(\lambda - \lambda_j) \frac{1}{r} - \frac{1}{r_e} \right) \right] \\ &+ \sum_{l=0}^{\infty} \sigma_{\infty, \ell}(\lambda_e) \left(\frac{r}{r_e} \right)^\ell P_\ell(\zeta) \\ &\text{for every } r \in [0, +\infty), \zeta \in [-1, 1] \text{ and } \phi \in [0, 2\pi), \quad (23) \end{aligned}$$

which is the reduced spherical field that coincides with the respective result in [12], recovered in the very simple and useful form (23) for the case of continuous consumption rates.

5 Discussion and conclusions

In this research, we developed a model to study nutrient distribution within a fixed oblate spheroidal tumour structure situated inside a finite confocal oblate spheroidal host microenvironment. The model assumes a fixed geometry and does not account for the dynamics of evolving tumour growth. We considered the diffusion of inhomogeneous nutrient concentration as the dominant process regulating the tumour's nutrient distribution within the surrounding medium. To create a more realistic model compared to existing ones, we assumed certain surface discontinuities in the nutrient flow across distinct boundaries.

The model was mathematically formulated as several interconnected boundary value problems for the nutrient field, involving elliptic-type Poisson's partial differential equations in each compartment of the avascular tumour. These were accompanied by appropriate Dirichlet or Neumann continuity conditions at the interfaces and specific nutrient supply from the external environment. These problems were solved incrementally, providing the internally consumed nutrient field throughout the oblate spheroidal tumour and the surrounding host environment in a concrete manner via an infinite series expansion using oblate spheroidal harmonic eigenfunctions.

This procedure offered insights into the distribution of the nutrient supply and consumption rates in a non-homogeneous tumour structure. In addition, an extensive discussion about the nutrient's critical values was included to complete the current analysis. The derived analytical forms were supported by a consistent numerical implementation of the results for nutrient distribution within the avascular tumour, presented through diagrams in both two-dimensional and three-dimensional regimes. The illustrations showed the expected behaviour of the nutrient, with its value decreasing as we move towards the tumour's necrotic core, thus ensuring our modelling technique avoids jump discontinuities at the layer boundaries. Finally, a convenient transformation was presented, which can easily adapt the model to a corresponding prolate spheroidal model. Our model also demonstrated a geometrical reduction to the special case of complete geometrical isotropy, where the tumour is considered spherical, thus, recovering existing results for the sphere.

In a biological system, nutrient distribution plays a critical role in determining cell phenotypes rather than being determined by them. The fixed nutrient distribution modelled here helps us to understand the potential locations and conditions under which different cell phenotypes might emerge based on nutrient availability.

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Author contributions P.V. was involved with the analytical methodology and wrote the main manuscript. P.P. was involved with the numerical implementation, prepared all figures. All authors reviewed the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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